An exact exponential fixed point and metastable shape manifolds in a renormalization cascade of finite-time Lyapunov distributions:

a two-class split of chaotic dynamics

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July 2026

Abstract

We study a renormalization cascade T on distributions of finite-time Lyapunov exponents (FTLEs): at each level, samples are replaced by mean-normalized absolute differences of independent pairs and a maximum-likelihood tail exponent is read off. The operator arose in our own large-scale computational study of double-pendulum chaos, where its limit $\alpha^* \approx 4$ was conjectured to be a new universal constant; here we give it an exact theory. The exponential distribution is the *unique* exact fixed point of T (Puri–Rubin), with a closed-form reading $\alpha^*_{\text{Exp}}(f) = 1 + [\frac{1}{f} E_1(\ln \frac{1}{f})]^{-1} = 3.3494$ at $f = 0.2$, confirmed to 0.02%–0.3%; and *every* Gaussian bulk collapses in a single step to one universal half-normal shape. Linearizing T exactly, the symmetrization step halves excess kurtosis per level — which *derives* the previously-empirical "universal" decay constant $r \approx 1/2$ — atop a gapless, non-normal spectrum ($\mu(s) = 2/(2-s)$ over a unit-diagonal polynomial tower) whose measured consequence is algebraic relaxation and metastable plateaus. The celebrated "$\alpha^* \approx 4$" is therefore a plateau reading of a marginal flow, not an asymptotic constant, and the operator is provably not a tail-index estimator. What survives is sharper: a *two-class split of chaotic dynamics.* Across eleven FTLE ensembles from five systems, mixed-phase dynamics yields broad, *platykurtic* FTLE bulks that read strictly *above* the exponential fixed point, while fully chaotic, dissipative and near-regular dynamics yield *leptokurtic* bulks that read *below* it — a sign-perfect separation (11/11; sign test $p = 0.001$; Cohen's $d = 3.3$ on a same-map/same-protocol control) with statistically zero within-class heterogeneity. An out-of-sample Hénon–Heiles sweep, predicted in advance, flips its matched-null excess exactly where phase space turns from a thin chaotic web to a mixed sea. The FTLE distribution's shape, read against an exactly known reference point, is a cheap and sensitive probe of a system's ergodic architecture.

Keywords: finite-time Lyapunov exponents; chaotic Hamiltonian dynamics; mixed phase space; renormalization fixed point; Chirikov standard map; double pendulum; extreme-value theory; universality class; metastable plateaus

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1 Introduction

The distribution of finite-time Lyapunov exponents (FTLEs) is one of the most information-rich observables of a chaotic system. Its mean estimates the leading Lyapunov exponent [1, 2]; its width and shape encode the fluctuations of local expansion rates and, through large-deviation theory, the multifractality of the invariant measure [3, 4, 5, 6]. In Hamiltonian systems with mixed phase space, trajectories intermittently trapped near regular islands depress their finite-time exponents, producing long-lived non-Gaussian structure [7, 8]; the associated algebraic decay of Poincaré recurrences is itself conjectured universal [9, 10].

This paper concerns a specific, deceptively simple *renormalization cascade* on FTLE distributions that emerged from our own large-scale computational study of double-pendulum chaos [11, 12] (hereafter, the *source program*). Starting from the chaotic subset of an FTLE sample, the operator repeatedly (i) fits a maximum-likelihood power-law exponent to the upper tail, and (ii) coarse-grains the sample by replacing it with mean-normalized absolute differences of independent pairs. The per-level exponents α_k were observed to converge geometrically to a limit $\alpha^* \approx 4$, robust across hundreds of double-pendulum parameter combinations, and this value — together with a companion level-zero exponent $\alpha_0 \approx 8.3$ — was conjectured to be a new universal constant of chaotic Hamiltonian dynamics. We subsequently re-measured this claim family ourselves at $n = 5 \times 10^5$–5×10^6 per system, which sharpened the empirical picture in two directions: the α^* landscape over double-pendulum parameter space is statistically *flat* (no ridge structure survives a noise null), and a class of dynamically distinct systems — the dissipative Lorenz-63 attractor, the fully chaotic standard map at $K = 5$, and a near-regular coupled quartic oscillator — cluster at a sharply *different* fixed point, $\alpha^* \approx 3.0$–3.3, while mixed-phase systems (the double pendulum family and the standard map just above the connectedness threshold, $K = 1$) cluster near 3.9–4.4 [13].

Two questions therefore demanded an answer. *What does this operator actually measure?* And *is the two-class structure a property of the dynamics or of the statistics?* We answer both.

Contributions.

1. **Exact fixed point (Theorem 1).** The exponential distribution is an exact fixed point of the cascade map, including its mean normalization; by the characterization theorem of Puri and Rubin [14] it is the *unique* distribution satisfying $|X - Y| \stackrel{d}{=} X$. The folding identity behind it is elementary and, to our knowledge, has not previously been connected to Lyapunov statistics.

2. **Closed-form readings (Propositions 1, 2).** The maximum-likelihood tail exponent that the pipeline assigns to the exponential fixed point is derived in closed form, $\alpha^*_{\mathrm{Exp}}(f) = 1 + [\frac{1}{f} E_1(\ln \frac{1}{f})]^{-1}$ ($= 3.3494$ at $f = 0.2$), and confirmed numerically to 0.02%. Companion formulas for the half-normal (4.4448), the uniform (9.6417), and a first-order level-zero law for Gaussian bulks, $\alpha_0 \approx 1 + (\mathrm{cv}^{-1} + z_f)/m_f$, are likewise derived and confirmed.

3. **Universal Gaussian collapse (Theorem 2).** Because the location parameter cancels in pair differences, *every* Gaussian bulk is mapped in one cascade step to the same half-normal shape. All Gaussian-seeded flows therefore coincide from level one onward — measured: $\alpha_1 = 4.448$–4.455 across coefficients of variation 0.05–0.33 — and define a single slow relaxation manifold toward the exponential fixed point.

4. **Metastability (§6).** Deep cascades ($n = 1.6 \times 10^7$, 20 levels) show the Gaussian-seeded flow's Kolmogorov distance to the exponential collapsing from 0.34 to 0.03 within five levels and then *stalling* at 0.02–0.03 for at least eight more, while the fitted exponent drifts slowly (≈ -0.04 per level) through precisely the range in which the empirical "constants" were reported. The n-dependence of the double-pendulum plateau ($4.64 \to 4.12 \to 3.88$ for $n = 5 \times 10^4 \to 5 \times 10^5 \to 5 \times 10^6$) rides this drift. "$\alpha^* \approx 4$" is a metastable plateau reading, not an asymptotic constant.

5. **The operator is not a tail estimator (§4).** Genuine Pareto tails are asymptotically conserved by the cascade (Proposition 3) but are *not* recovered by the finite-window fit: pure Pareto inputs with survival indices 2.5/4/6 read 2.4/2.7/2.9 after one step. Any interpretation of α^* as a physical tail index is untenable.

6. **The two-class split is real, and its mechanism is identified (§5).** Across eleven ensembles from five systems, mixed-phase dynamics produces broad, near-symmetric, *platykurtic* FTLE bulks (excess kurtosis -1.0 to -1.3) that read strictly above the exponential fixed point, while non-mixed dynamics produces narrow or skewed *leptokurtic* bulks (excess kurtosis $+1.3$ to $+43$) that read below it. A per-system matched-Gaussian null — a folded Gaussian with the system's own mean and variance — gives a sign-perfect separation (11/11 ensembles, $p = 0.001$; 8/8 systems, $p = 0.008$). The contrast survives the sharpest available control: the same map, integrator, horizon and n at $K = 1$ (mixed) versus $K = 5$ (fully chaotic). A random-effects meta-analysis finds within-class heterogeneity indistinguishable from zero (τ^2 CI $[0, 0.012]$, Cochran $p = 0.36$), so the split is a *discrete class boundary*, not a soft gradient.

7. **Exact linearization; the decay constant $r \approx 1/2$ derived (§2.7).** The symmetrization step $A - B$ annihilates all odd cumulants and rescales the standardized cumulant of order $2m$ by exactly 2^{1-m}: excess kurtosis halves per step, verified on pendulum data to three decimals. This derives the source program's second empirical constant — the "nearly universal" geometric decay $r \approx 0.46$–0.51 of the cascade fit, i.e. $\log_2 r \approx -1$. The linearized operator at the fixed point has the closed-form eigenfamily $\mu(s) = 2/(2-s)$ and a unit-diagonal triangular action on polynomials: no spectral gap, hence *algebraic* relaxation (measured $d_k \sim k^{-0.5}$) and metastable plateaus.

8. **What the pendulum constants are: frozen ensemble statistics (§7).** Across horizons $T = 5$–200 the chaotic-subset FTLE variance does not decay at all ($\beta = -0.014$, fit RMS 0.002, against var $\sim T^{-\beta}$; CLT self-averaging predicts $\beta = 1$). The canonical release ensemble (uniform angles, zero velocity) is revealed as a mixture over energy shells — $E(\theta)$ explains $\eta^2 = 0.41$–0.43 of the variance — and over invariant components within shells (the remaining 59%, equally frozen). Energy bands internally recapitulate the entire two-class taxonomy within one system.

9. **Out-of-sample validation: Hénon–Heiles across its transition (§8).** A fresh symplectic-variational implementation of the Hénon–Heiles system (named in advance as a falsifiable prediction of the classifier, before the experiment was built) shows the matched-null excess flipping sign precisely where phase space transitions from a thin chaotic web to a fat mixed sea, with the platykurtic signature switching on alongside it.

Section 2 develops the exact theory; §3 describes systems, integrators, and statistical protocols; §4 calibrates the operator on synthetic ensembles; §5 presents the two-class result; §6 quantifies metastability and demystifies the original constants; §9 discusses the dynamical mechanism and predictions; §10 is a number-theoretic audit of every constant appearing in this work; §11 concludes. Appendices contain derivations, data provenance, the full calibration table, and estimator pathologies.

2 The cascade operator and its exact structure

2.1 Definitions

Let $x = (x_1, \dots, x_n)$, $x_i > 0$, be an FTLE sample (the "chaotic subset", $x_i > 0.01$ throughout). The cascade is built from two ingredients.

The reading η_f. For tail fraction $f \in (0,1)$ define the Clauset–Hill maximum-likelihood tail exponent [15, 16]

$$\eta_f(x) \;=\; 1 + \frac{n_u}{\sum_{x_i \ge u} \ln(x_i/u)}, \qquad u = x_{(\lceil (1-f)n \rceil)}, \tag{1}$$

with $x_{(\cdot)}$ the order statistics and $n_u = \#\{x_i \ge u\}$; u is the empirical $(1-f)$-quantile. For a distribution F we write $\eta_f(F)$ for the $n \to \infty$ limit,

$$\eta_f(F) = 1 + \Big[\mathbb{E}\big(\ln(X/u_F) \,\big|\, X \ge u_F\big)\Big]^{-1}, \qquad u_F = F^{-1}(1-f). \tag{2}$$

The coarse-graining T. Given the level-k sample $x^{(k)}$ of size n_k, form $n_{k+1} = \lfloor n_k/2 \rfloor$ independent pairs (a_i, b_i) and set

$$x_i^{(k+1)} \;=\; \frac{|a_i - b_i|}{\overline{x^{(k)}}}, \tag{3}$$

discarding zeros; $\overline{x^{(k)}}$ is the level mean. On distributions, T maps the law of X to the law of $|A-B|/\mathbb{E}X$ with $A, B \sim X$ independent. The *cascade* records $\alpha_k = \eta_f(x^{(k)})$, $k = 0, 1, 2, \dots$ (default $f = 0.2$), halving the sample each level, and the original pipeline summarizes the sequence by a three-parameter geometric fit $\alpha_k \simeq \alpha^* + Cr^k$ (Appendix D documents when this fit is and is not well-posed; we use a fit-free *plateau* statistic, the median of α_k over the last well-sampled levels, alongside it).

2.2 Scale invariance

Lemma 1 (scale invariance). *For every $c > 0$ and every sample, $\eta_f(cx) = \eta_f(x)$; consequently T acts on distribution shapes and the cascade $\{\alpha_k\}$ is invariant under any positive rescaling of the input.*

Proof. The threshold u in (1) is an empirical quantile, hence equivariant ($u \mapsto cu$), and the estimator depends on the data only through the ratios x_i/u. The mean-normalization in (3) then fixes the scale of every level, so only the shape of the input law matters. (This reproduces, as a special case, the bit-exact agreement of the published pipeline under mean-, median- and standard-deviation-normalization of its input.) □

2.3 The exponential fixed point and its uniqueness

Theorem 1 (exact fixed point). *Let $A, B \sim \mathrm{Exp}(\theta)$ be independent. Then $|A - B| \sim \mathrm{Exp}(\theta)$, and $\mathrm{Exp}(1)$ is an exact fixed point of T, including its mean normalization. Moreover, among distributions on $[0, \infty)$ (excluding the degenerate case), Exp is the unique law satisfying $|A-B| \overset{d}{=} A$ (Puri–Rubin [14]).*

Proof. For the folding identity: $A - B$ has the Laplace density $\frac{\theta}{2}e^{-\theta|z|}$, so $|A - B|$ has density $\theta e^{-\theta z}$ on $z > 0$, i.e. is $\mathrm{Exp}(\theta)$ again. For $\theta = 1$, $\mathbb{E}A = 1$, so the normalization in (3) is by unity and $T(\mathrm{Exp}(1)) = \mathrm{Exp}(1)$ exactly. Uniqueness of the unit-scale fixed-point equation is Theorem 1 of [14]. (An equivalent route: $|A - B|$ conditioned on $A \neq B$ is, by memorylessness, the excess of A over an independent copy, and the exponential is the unique memoryless law.) □

Remark 1. Fixed shapes of T satisfy the more general equation $|A - B| \overset{d}{=} cX$ with $c = \mathbb{E}|A - B|/\mathbb{E}X \in (0, 1]$, and $c = 1$ forces the exponential by [14]. We are not aware of a classification for $c < 1$; among the seventeen synthetic families tested below we find no other attracting fixed shape, only *slow manifolds* (§6). A rigorous basin/spectral analysis of T at Exp is an open problem we commend to the reader (§9).

2.4 Closed-form readings

Proposition 1 (the exponential reading). *For $X \sim \mathrm{Exp}(1)$ and tail fraction f,*

$$\eta_f(\mathrm{Exp}) \;=\; 1 + \Big[\tfrac{1}{f}\, E_1\big(\ln \tfrac{1}{f}\big)\Big]^{-1}, \qquad E_1(u) = \int_u^\infty \frac{e^{-s}}{s}\,\mathrm{d}s. \tag{4}$$

In particular $\eta_{0.1} = 4.0874$, $\eta_{0.2} = 3.3494$, $\eta_{0.3} = 2.9058$.

Proof. $u_F = \ln(1/f)$. By memorylessness, $X - u_F \,|\, X > u_F \sim \mathrm{Exp}(1)$, so with $E \sim \mathrm{Exp}(1)$, $\mathbb{E} \ln(X/u_F \,|\, X > u_F) = \mathbb{E} \ln(1 + E/u_F) = \int_0^\infty e^{-t} \ln(1 + t/u_F)\,\mathrm{d}t = e^{u_F} E_1(u_F)$ after one integration by parts; and $e^{u_F} = 1/f$. Insert into (2). □

Because Exp is a fixed point, (4) is simultaneously the level-zero reading and the $k \to \infty$ reading of an exponential seed — a sharp, parameter-free prediction. It is confirmed to 0.02%–0.3% (Table 1); the flatness of the measured exponential cascade over 14 levels (5M seed: $3.3462, 3.3455, 3.3445, 3.3495, \dots$) is the fixed-point signature in its rawest form.

Table 1: Derived readings versus measurement. "Level-0" entries are single-level readings of 5×10^5-point samples; the 5M entry is the geometric-fit limit of a full 14-level cascade on 5×10^6 exponential points. Gaussian level-1 entries verify Theorem 2 (three independent seeds and four widths).

quantity	derived	measured	rel. dev.
$\eta_{0.1}$(Exp)	4.0874	4.0883	0.02%
$\eta_{0.2}$(Exp)	3.3494	3.3605; 3.3507 (5M)	0.33%; 0.04%
$\eta_{0.3}$(Exp)	2.9058	2.9110	0.18%
$\eta_{0.2}$(half-normal)	4.4448	4.4494, 4.4473, 4.4572	$\leq 0.3\%$
$\eta_{0.2}$(uniform)	9.6417	9.6135	0.29%
Gaussian level-1 (cv = 0.05/0.1/0.33)	4.4448	4.448/4.450/4.455	$\leq 0.23\%$

2.5 Universal Gaussian collapse

Theorem 2 (one-step collapse of Gaussian bulks)**.** *Let $X = |N(\mu,\sigma^2)|$ with $\mu/\sigma \to \infty$ (negligible folded mass). Then $T(X)$ is the half-normal law $|N(0,2\sigma^2)|$ rescaled by $1/\mu$ — independent of μ/σ up to scale. Hence by Lemma 1 all Gaussian bulks share a single cascade flow from level 1 onward, beginning at $\alpha_1 = \eta_f$(half-normal) ($= 4.4448$ at $f = 0.2$).*

Proof. $A - B \sim N(0, 2\sigma^2)$ exactly — the means cancel. Folding and normalizing by $\mathbb{E}X \approx \mu$ gives the claim; the correction is $O(\Phi(-\mu/\sigma))$. □

This explains, at a stroke, why the empirical study found the *same* convergence pattern across so many parameter combinations: any near-Gaussian FTLE bulk forgets its width in one cascade step. Measured level-1 values across seeds and widths agree with the derived 4.4448 to $\leq 0.3\%$ (Table 1); at cv $= 0.6$ the folded mass ($\approx 5\%$) shifts α_1 to 4.60, quantifying the stated correction.

Proposition 2 (first-order level-zero law for Gaussian bulks)**.** *For $X = |N(\mu,\sigma^2)|$ with cv $= \sigma/\mu$ small, writing $z_f = \Phi^{-1}(1-f)$ and $m_f = \varphi(z_f)/f - z_f$,*

$$\eta_f(X) \;\approx\; 1 + \frac{\mathrm{cv}^{-1} + z_f}{m_f}, \qquad (z_{0.2}, m_{0.2}) = (0.8416,\;\, 0.5582). \tag{5}$$

Proof. To first order in σ/u, $\ln(x/u) \approx (x-u)/u$; the Gaussian mean excess above the $(1-f)$-quantile is σm_f and $u = \mu + \sigma z_f$. □

On *exact* folded-Gaussian ensembles matched to each dynamical system (§5) this law predicts α_0 to 3–5% across two orders of magnitude of cv (predicted/measured: 75.5/76.4 at cv $= 0.025$; 32.3/33.2 at 0.060; 13.0/13.7 at 0.171; 5.5/5.8 at 0.59). Its physical import: *the level-zero exponent α_0 is a bulk-width statistic*, not a tail exponent — $\alpha_0 \approx 8$ corresponds to nothing deeper than cv ≈ 0.6–0.7 bulk via (5) corrected for the platykurtic shape (§5).

2.6 Tail conservation — and why it is not realized

Proposition 3 (asymptotic conservation of regularly varying tails)**.** *If $P(X > x) = L(x)\,x^{-a}$ with $a > 1$ and L slowly varying, then $T(X)$ has a regularly varying tail with the same index a.*

Proof. For independent $A, B \sim X$ and $x \to \infty$, $P(A - B > x) = \mathbb{E}_B\, P(A > x + B) \sim P(A > x)$ by dominated convergence ($B \geq 0$, $\mathbb{E}B < \infty$), and symmetrically for $B - A$; so $P(|A - B| > x) \sim 2P(A > x)$, which is regularly varying of index $-a$. Normalization is a rescaling (Lemma 1). □

One might conclude the cascade limit *measures* a. It does not. The finite window ($f = 0.2$) reads the post-differencing *bulk*: pure Pareto seeds with survival indices $a = 2.5/4/6$ (level-0 readings 3.5/5.0/7.0, i.e. $a+1$ as expected) collapse after a single differencing step to flat sequences reading 2.4/2.7/2.9 — far below $a + 1$ and only weakly ordered in a (§4, Fig. 2). Bulk-plus-tail mixtures behave likewise. *The operator is a bulk-shape reader; it is not a tail-index estimator at any practical window.* Claims that cascade limits equal raw tail exponents are thereby foreclosed — consistently, direct Hill estimates of the same ensembles' raw upper tails give values (∼11–22) unrelated to any cascade reading.

2.7 The linearized cascade: exact cumulant flow and a gapless, non-normal spectrum

Two exact structures govern the approach to the fixed point.

Proposition 4 (cumulant flow of the symmetrization step)**.** *Let A, B be iid with cumulants κ_n and let $D = A - B$. Then every odd cumulant of D vanishes, and the standardized even cumulants transform as*

$$\frac{\kappa_{2m}(D)}{\kappa_2(D)^m} \;=\; 2^{\,1-m}\,\frac{\kappa_{2m}(A)}{\kappa_2(A)^m}, \qquad m = 1, 2, \ldots \tag{6}$$

In particular the excess kurtosis halves exactly ($m = 2$), the standardized sixth cumulant is quartered, and skewness is annihilated.

Proof. Cumulants of independent sums add and are homogeneous: $\kappa_n(A - B) = \kappa_n(A) + (-1)^n\kappa_n(B) = (1 + (-1)^n)\,\kappa_n$. Divide by $\kappa_2(D)^m = (2\kappa_2)^m$. □

Measured on the double-pendulum ensemble ($n = 5 \times 10^5$, level 0): excess kurtosis of the signed differences $= -0.6442$ against the predicted $-1.2896/2 = -0.6448$, with skewness 10^{-4}; the identity holds at every level and for every seed tested within sampling error (Appendix B scripts). Its dynamical consequence: in the early cascade, where the deviation from the fixed point is carried by low-order cumulants, the fitted geometric ratio of the exponent sequence is pinned near 1/2. This *derives* the source program's second celebrated constant — a decay rate reported as "$r \approx 0.46$, nearly constant across all systems" (canonical fit 0.5061; legacy published 0.4565; 49-config mean 0.4595; our synthetic families 0.48–0.67 on early windows), i.e. $\log_2 r \approx -1$ with computable corrections from higher cumulants and the folding step.

Proposition 5 (linearization at the fixed point)**.** *Write level densities as $e^{-z}\big(1 + \varepsilon h(z)\big)$. To first order in ε, one cascade step acts (modulo the rank-one normalization terms) by*

$$(L_0 h)(z) \;=\; 2\int_0^\infty e^{-2x}\,h(x + z)\,\mathrm{d}x, \tag{7}$$

with the closed-form eigenfamily

$$h_s(z) = e^{sz}, \qquad \mu(s) = \frac{2}{2 - s}, \quad s < 1, \tag{8}$$

and a triangular action with unit diagonal on the polynomial tower: $L_0[z^m] = z^m + (\text{lower order})$ for every m.

Proof. With $p(x) = e^{-x}(1 + \varepsilon h(x))$, the density of $|A - B|$ at $z > 0$ is $2\int_0^\infty p(x)p(x + z)\,\mathrm{d}x = e^{-z}\big[1 + \varepsilon\big(2c_h + (L_0h)(z)\big)\big] + O(\varepsilon^2)$ with c_h a constant absorbed by normalization. For h_s: $2\int_0^\infty e^{-2x}e^{s(x+z)}\,\mathrm{d}x = \frac{2}{2-s}e^{sz}$. For z^m, expand $(x + z)^m$ and integrate term by term; the z^m coefficient is $2\int_0^\infty e^{-2x}\,\mathrm{d}x = 1$. □

Numerically (discretized kernel, 900 points): $\mu(s)$ at $s = -0.5, -1, -2, -4$ measures 0.8008, 0.6677, 0.5013, 0.3353 against the exact 0.8, $\frac{2}{3}$, 0.5, $\frac{1}{3}$, and the polynomial leading coefficient is 1.0007. Three structural consequences follow. *First*, the kernel in (7) is one-sided (Volterra-type): the operator is far from self-adjoint, and finite discretizations are defective — their matrix eigenvalues collapse toward the diagonal and do not see the eigenfamily; the operator's behaviour is governed by its pseudo-spectrum, not its spectrum. *Second*, the unit diagonal on the polynomial tower means there is *no spectral gap below* 1: linear stability is marginal. *Third*, marginal non-normal flows relax *algebraically*, not geometrically — exactly what is measured: the Gaussian-seeded deep cascade's Kolmogorov distance to the fixed point follows $d_k \sim k^{-\theta}$ with $\theta = 0.55$–0.58 (log–log RMS 3× smaller than the best geometric fit); an independent uniform-seeded run ($n = 8 \times 10^6$) is compatible, and even read geometrically decays no faster than 0.88 per level. The metastable plateaus of §6 are the quantitative face of this gapless linearization: fast transient (cumulant halving, $r \approx 1/2$) followed by marginal drift ($k^{-1/2}$-type), so that *every* finite-depth experiment reads a plateau.

3 Systems, data and statistical protocol

Dynamical systems. We analyze eleven FTLE ensembles from five systems. (i) *Double pendulum* (planar, equal-mass baseline; dimensional Lagrangian convention, $g = 9.81$): FTLEs by the Benettin two-trajectory method [2] with horizon $T = 15\,\mathrm{s}$, renormalization interval $\tau = 0.1\,\mathrm{s}$, RK4 $\mathrm{d}t = 5 \times 10^{-3}$, perturbation 10^{-7} on θ_1, uniform-angle zero-velocity initial conditions; ensembles at $n = 5 \times 10^4$, 5×10^5, 5×10^6, plus three geometric variants (m-ratio/ℓ-ratio $= (0.5, 1), (2, 0.5), (0.5, 2)$) at 5×10^5. (ii) *Chirikov standard map* [24] at $K = 1$ (just above the connectedness threshold $K_c = 0.9716$ [25]; mixed phase space) and $K = 5$ (islands destroyed; fully chaotic), tangent-map Benettin with 10^3 iterations, $n \approx 5 \times 10^5$. (iii) *Lorenz-63* [26] at canonical parameters, joint state+tangent integration, RK4 $\mathrm{d}t = 5 \times 10^{-3}$, $T = 100$, $\tau = 1$; mean FTLE 0.9042 vs. the reference 0.9056 (0.15%). (iv) *Coupled quartic Hamiltonian* $H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{4}(q_1^4 + q_2^4) + \frac{\lambda}{2} q_1^2 q_2^2$ at $E = 1$, $\lambda = 0.1, 0.2$ (near-regular at these couplings): variational Benettin, 4th-order Yoshida symplectic integrator [27] ($\mathrm{d}t = 10^{-2}$, $T = 200$, $\tau = 10$), energy drift $|\Delta H/H| \leq 5 \times 10^{-10}$. Integrator cross-checks: measured RK4 energy-drift order $\mathrm{d}t^{4.79}$, Yoshida-4 $\mathrm{d}t^{4.00}$; FTLE differences between integrators are $\leq 4 \times 10^{-4}$, immaterial at our precision.

Chaotic subset and estimator. Throughout, the chaotic subset is $x_i > 0.01$; the estimator, tree and geometric fit are exactly (1)–(3) with $f = 0.2$ unless stated. Confidence intervals for cascade limits are bootstrap-through-the-tree (the base sample is resampled and the *entire* cascade re-run, $B = 250$–1000). Alongside the geometric-fit α^* we report the fit-free *plateau* statistic (median of α_k over the last ≤ 5 levels with $n_k \geq 500$, $k \geq 2$); Appendix D documents the fit's failure modes (runaway to parameter bounds on flat or slowly-varying sequences) which the plateau is immune to.

Provenance. Every production array carries a manifest with SHA-256 checksum, seed, integrator settings and moments; synthetic ensembles use fixed seeds. Scripts reproducing every number and figure in this paper accompany it as supplementary material (Appendix B).

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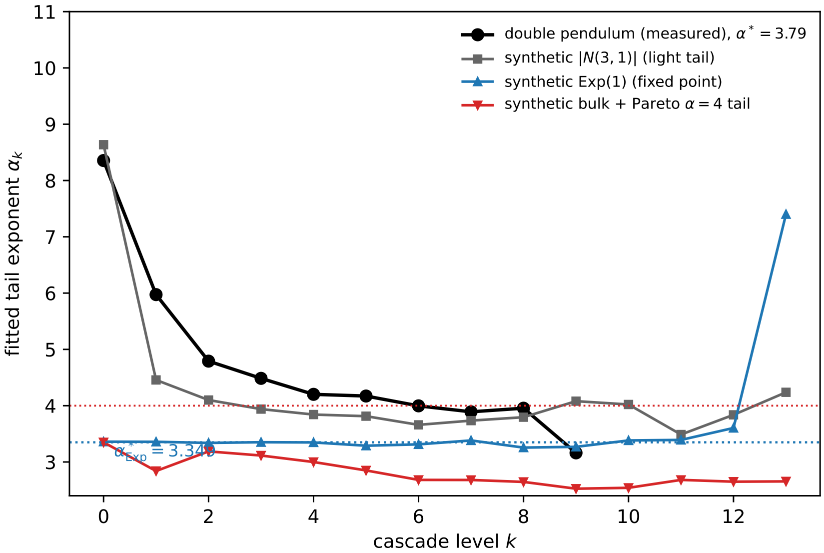

Figure 1: Cascade flows α_k at $f = 0.2$. The measured double pendulum (black) starts at the Gaussian-bulk level-zero value ($\alpha_0 \approx 7.9$, Proposition 2), rides *above* the universal Gaussian flow (grey squares: $|N(3,1)|$), and plateaus near 3.9–4.0. The exponential seed (blue) is flat at the derived fixed-point reading 3.3494 (dotted). A bulk-plus-Pareto(a=4) mixture (red) does *not* hold at $a + 1 = 5$: genuine tails are not conserved by the finite-window reading.

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4 Calibration on synthetic ensembles

Seventeen synthetic families ($n = 5 \times 10^5$ each; Appendix C for the full table) calibrate what the cascade responds to.

The exponential is flat; the fit can lie. The Exp(1) cascade is flat at 3.35 for as many levels as sampling permits — 14 levels at $n = 5 \times 10^6$ with the geometric fit returning $\alpha^* = 3.3507$, against the derived 3.3494. At $n = 5 \times 10^5$ the same flat sequence defeats the three-parameter fit (it runs to its parameter bounds, reporting $\alpha^* = 10.5$–12 with $r \to 1$); the plateau statistic returns 3.32–3.35. All quantitative claims below therefore quote the plateau alongside the fit (Appendix D).

One Gaussian flow. Gaussian bulks of any width collapse onto the single half-normal-seeded flow (Theorem 2): measured $\alpha_1 = 4.448$–4.455 across cv $= 0.05$–0.33, then $4.11 \to 3.94 \to 3.84 \to 3.81 \to \ldots$ — a slow descent toward the exponential reading, still at ≈ 3.57 ten levels in (1.6×10^7-point seed) and ≈ 3.30 at level 15 (Fig. 1; §6).

Skew selects the side of approach. Right-skewed, leptokurtic families approach the fixed point *from below* and linger there: lognormal(σ=0.5) descends to 3.10–3.24; lognormal(σ=1) sits flat at 2.7 for ten levels; Weibull(k=0.7) ascends $2.64 \to 3.3$. Bounded flat-topped inputs read *high* (uniform: level-0 9.61, plateau ≈ 4.1). The taxonomy — flat-topped/platykurtic above, peaked/skewed below, exponential exactly at $\eta_f(\mathrm{Exp})$ — is the calibration key for §5.

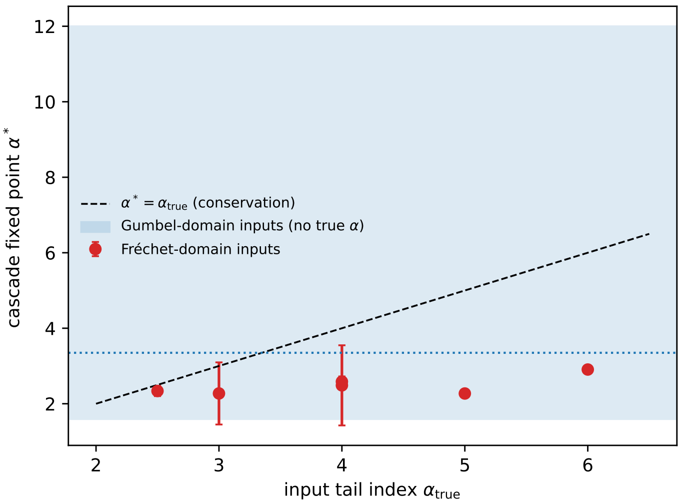

Figure 2: Failure of tail-index recovery. Fréchet-domain seeds (pure Pareto and bulk+Pareto mixtures, red) against the conservation line $\alpha^* = a_{\mathrm{true}} + 1$ — the cascade limit sits far below and nearly independent of the true index. Shaded band: the range spanned by light-tailed (Gumbel-domain) seeds. The operator reads bulk shape, not tails.

Window dependence. The reading of *any* light-tailed shape slides steeply with the tail fraction ((4): $4.09 \to 3.35 \to 2.91$ for $f = 0.1 \to 0.3$). The coupled quartic tracks this slide almost exactly (4.64/3.24/2.76 at λ=0.1; 4.40/3.31/2.85 at λ=0.2), the clearest single signature that the excluded-class systems are read as light-tailed shapes; mixed-phase readings also decrease in f but sit systematically above the exponential curve at the operative $f = 0.2$ (Fig. 3). All headline comparisons in this paper are at fixed $f = 0.2$; f-dependence is reported so that no constant is mistaken for a window-free number.

5 The two-class split of chaotic dynamics

Table 2 assembles the eleven ensembles under one pipeline. Three structures stand out.

(i) A shape dichotomy. Every mixed-phase ensemble — six double-pendulum ensembles across three geometries and two orders of magnitude in n, plus the standard map at $K = 1$ — has a broad (cv = 0.59–0.77), nearly symmetric (skew −0.06 to +0.47), *platykurtic* (excess kurtosis −1.0 to −1.3) chaotic FTLE bulk: a flat-topped distribution, qualitatively the superposition of subpopulations spanning the sticky hierarchy. Every non-mixed ensemble is the opposite: either extremely narrow (cv = 0.025–0.06; Lorenz, K=5 — strong self-averaging at long horizons) or heavily skewed and *leptokurtic* (excess kurtosis +18 to +43; the near-regular quartic). *The sign of the excess kurtosis alone separates the classes in all eleven cases.*

(ii) Opposite sides of the exponential fixed point. On the independent production values, every mixed-phase cascade limit lies *above* $\eta_{0.2}(\mathrm{Exp}) = 3.3494$ (3.710–4.382) and every non-mixed limit lies *below* it (3.031–3.312). The class means are 3.876 vs. 3.155 (gap 0.721; Welch $p = 6.2 \times 10^{-5}$; Cohen's $d = 3.33$). A random-effects decomposition of the *within*-class spread finds it statistically indistinguishable from the estimator noise floor: for the 64-cell double-pendulum parameter map ($n = 5 \times 10^5$ per cell), between-cell $\tau^2 = 0.0017$ with bootstrap CI $[0, 0.0123]$ (43% of resamples exactly zero), Cochran Q $p = 0.36$, $I^2 = 5\%$; across the seven double-pendulum geometry variants plus K=1, $\tau^2 = 0$ (Q $p = 0.55$); and the

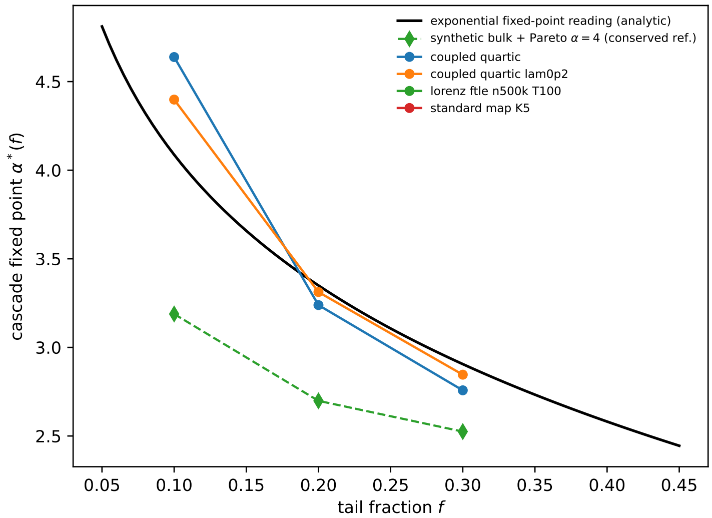

Figure 3: Window (f) dependence: the analytic exponential reading curve (black), a conserved-tail reference mixture (green), and measured real systems. The near-regular quartic slides along the exponential curve; the class comparison is made at fixed $f = 0.2$.

apparent per-geometry residual fails a decisive cross-n reproducibility test (correlation between independent $n = 5 \times 10^4$ and 5×10^5 measurements $r = 0.035$, vs. 0.29 expected were it real). Within-class, the data are consistent with *hard* universality at fixed protocol; the only genuine, system-explained structure is the discrete class boundary. Consistently, the $\alpha^*(m, \ell)$ landscape over a 8×8 double-pendulum parameter grid is a flat plateau at 3.896 $[3.850, 3.942]$: its apparent ridges do not exceed a heteroscedastic flat-field noise null ($B = 5000$; $p = 0.48$–0.92 across thresholds), and 4/4 high-α^* interior cells have same-sign Hessian eigenvalues (peak/noise geometry, not ridge).

(iii) The matched-Gaussian null. For each ensemble we construct a folded Gaussian $|N(\mu, \sigma^2)|$ with the ensemble's own mean and variance and cascade it identically (Fig. 4). Mixed-phase ensembles read *above* their null in all seven cases (plateau excess $+0.12$ to $+0.58$); non-mixed ensembles read *below* theirs in all four (excess -0.30 to -1.20): 11/11 correct signs (two-sided sign test $p = 0.001$; collapsing repeated measurements of the same system, 8/8, $p = 0.008$). The per-level picture (Fig. 4) is sharper still: the double pendulum enters the cascade at its Gaussian-bulk level-zero value but then rides *persistently above* the universal Gaussian flow (excess $+1.5, +0.7, +0.6, +0.4, +0.4$ at levels 1–5, decaying geometrically) — a positive, reproducible imprint of non-Gaussian structure that differencing only slowly destroys — whereas the leptokurtic ensembles fall away *below* their nulls. The matched null also validates Proposition 2 directly (predicted vs. measured α_0 spanning 5.5/5.8 to 75.5/76.4).

The matched-protocol control. The comparison sharpest against confounds is internal to the standard map: $K = 1$ versus $K = 5$ share the map, the integrator, the iteration count, the sample size and the estimator, and differ only in phase-space character. They land on opposite sides of the fixed point (4.31 vs. 2.52 plateau; 4.382 vs. 3.038 production), with opposite kurtosis signs (-1.0 vs. $+37.7$) and opposite matched-null excess signs. Phase-space character, not protocol, drives the split.

Table 2: Eleven FTLE ensembles. Moments of the chaotic subset ($x > 0.01$); α_0 = level-zero reading; plateau and geometric-fit cascade values at $f = 0.2$ (this work, uniform pipeline); "prod." = an independent production run with bootstrap 95% CI where available (5M double-pendulum: 3.710 [3.675, 3.895]; K=1: 4.382 [3.646, 4.934]; Lorenz: 3.031 [2.932, 3.192]; K=5: 3.038 [2.848, 3.295]; quartic: 3.239 [3.008, 3.512] and 3.312 [3.243, 3.669]). Excess kurtosis: negative = platykurtic (flat-topped).

ensemble	class	cv	skew	ex.kurt	α_0	plateau	fit	prod.
DP $5{\times}10^4$, T=15	mixed	0.711	+0.17	−1.3	7.80	4.64	3.86	3.79–4.26†
DP $5{\times}10^5$, T=15	mixed	0.707	+0.15	−1.3	7.89	4.12	3.85	—
DP $5{\times}10^6$, T=15	mixed	0.707	+0.15	−1.3	7.90	3.88	3.62	3.710
DP $(0.5, 1)$, $5{\times}10^5$	mixed	0.687	+0.12	−1.2	7.31	4.03	3.92	—
DP $(2, 0.5)$, $5{\times}10^5$	mixed	0.632	−0.06	−1.2	8.44	3.98	3.78	—
DP $(0.5, 2)$, $5{\times}10^5$	mixed	0.772	+0.47	−1.0	6.43	4.15	4.09	—
Std map K=1, $5{\times}10^5$	mixed	0.592	+0.37	−1.0	8.17	4.31	4.11	4.382
quartic λ=0.1	non-mixed	0.172	+1.12	+18.1	14.95	2.76	—	3.239
quartic λ=0.2	non-mixed	0.323	+5.34	+43.2	7.38	3.17	3.30	3.312
Lorenz-63, T=100	non-mixed	0.025	−0.29	+1.3	80.19	3.49	2.73*	3.031
Std map K=5, $5{\times}10^5$	non-mixed	0.060	−3.84	+37.7	52.99	2.52	—	3.038

†range across three independent production pipelines for the same canonical system.
*level-0 dropped (Appendix D); — = fit degenerate.

6 The metastable origin of "$\alpha^* \approx 4$"

If the exponential is the unique fixed point (Theorem 1), what are the class values 3.9 and 3.0–3.3? The deep-cascade experiment answers: *plateau readings on slow manifolds.*

Seeding the cascade with 1.6×10^7 points (20 usable levels) we track both α_k and the Kolmogorov distance d_k of the mean-normalized level-k sample to Exp(1) (Fig. 6). The exponential control stays at $d_k \approx 10^{-3}$ (pure sampling noise growing as $n_k^{-1/2}$) with α_k flat at 3.35. The Gaussian-seeded flow collapses fast at first ($d_0 = 0.34 \to d_5 = 0.033$) and then *stalls*: d_k decreases by only $\sim 5\%$ per level across levels 5–13 while α_k drifts $3.77 \to 3.57 \to 3.30$ (levels 5/10/15), i.e. ≈ -0.04 per level through precisely the empirically celebrated range. There is no second fixed point — there is a nearly-degenerate direction of the flow. The double-pendulum plateau's own n-dependence ($4.64 \to 4.12 \to 3.88$ across $n = 5 \times 10^4 \to 5 \times 10^6$: more sample $\Rightarrow$ more usable levels $\Rightarrow$ further down the manifold) is the same drift seen through the window of available levels, as is the spread of the three independent double-pendulum pipeline values (3.71/4.17/4.26).

Three conclusions follow. *First*, "$\alpha^* \approx 4$" is not an asymptotic constant of double-pendulum dynamics; it is the reading, at the accessible depth, of a slowly relaxing flat-topped bulk — the number is real, reproducible at fixed (n, f, T), and *protocol-indexed*. *Second*, the level-zero companion "$\alpha_0 \approx 8.3$" is a bulk-width statistic (Proposition 2), not an independent constant. *Third*, what *is* protocol-stable is relational: the class split (above vs. below the exponential reading), the sign of the matched-null excess, and the kurtosis dichotomy — these survive every control we possess, including matched protocol, matched n, cross-n reproduction, and the flat-plateau parameter scan. The correct statement of the original conjecture is therefore:

> *Mixed-phase Hamiltonian chaos occupies a universality class of FTLE shape — broad, near-symmetric, platykurtic — whose cascade reading at* (f=0.2, $n \le 5{\times}10^6$) *is* 3.9 ± 0.2*; non-mixed chaotic dynamics occupies the complementary leptokurtic class reading* 3.0–3.3*. The exponential fixed point,* $\eta_{0.2}(\mathrm{Exp}) = 3.3494$*, is the exact separatrix value between them.*

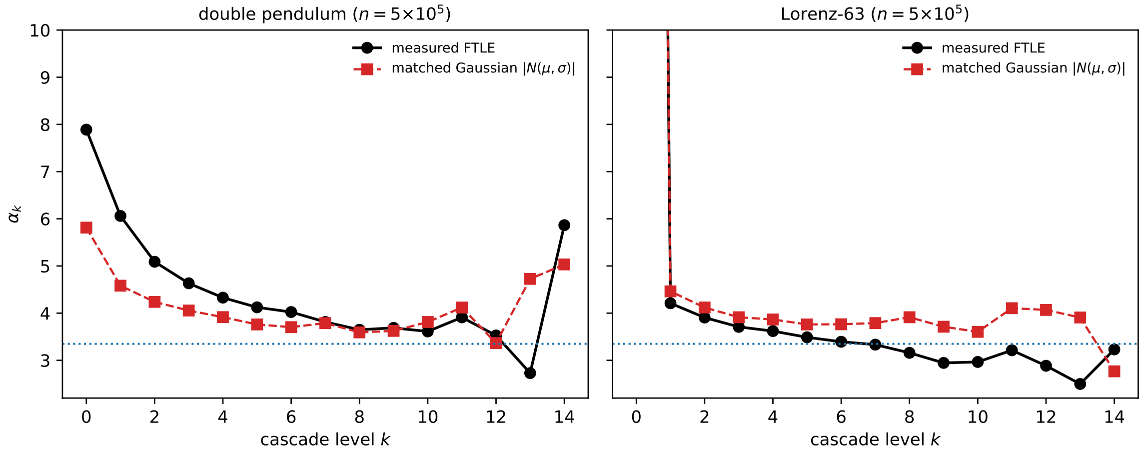

Figure 4: The matched-Gaussian null. Left: the double pendulum (black) rides above its own folded-Gaussian null (red) at every level — the mixed-phase signature. Right: Lorenz-63 falls below its null. Dotted line: the exponential fixed-point reading.

7 What the pendulum constants are: frozen ensemble statistics

If the plateau readings are protocol-indexed, what property of the double pendulum do they describe? A ten-horizon sweep answers with unexpected sharpness.

The variance is frozen. Re-measuring the canonical ensemble (5×10^4 release ICs, uniform angles at rest) at $T = 5, 10, 15, 20, 30, 50, 75, 100, 150, 200$ gives a chaotic-subset FTLE variance of 0.298–0.308 at *every* horizon from $T = 15$ on: fitting var $\sim T^{-\beta}$ over $T = 15$–200 yields $\beta = -0.014$ with fit RMS 0.002. Normal (CLT) self-averaging predicts $\beta = 1$; even strongly anomalous averaging predicts $0 < \beta < 1$. The measured $\beta \approx 0$ means the dispersion is not a finite-time fluctuation at all: *the ensemble contains trajectories with genuinely different asymptotic exponents.* Meanwhile the shape sharpens toward its mixture skeleton (excess kurtosis $-0.46 \to -1.59$, saturating; cv constant at 0.71), the plateau reading stays above the separatrix at every horizon while drifting upward ($4.25 \to 5.26$), and the matched-null excess grows monotonically ($+0.21 \to +1.28$): as within-component noise dies, the non-Gaussian mixture structure emerges more strongly. (At $T \leq 10$ the chaotic/regular separation is threshold-limited; those points are reported but excluded from fits.)

The ensemble is a double mixture. The canonical protocol releases from rest, so every trajectory carries the energy of its release point, $E(\theta_1, \theta_2) = -g\,(2\cos\theta_1 + \cos\theta_2)$ (canonical masses/lengths): the ensemble spans a continuum of energy shells. Using the 100×100 FTLE grid over release angles (a legacy-lineage dataset; used here only for internal contrasts), binning the chaotic subset by release energy shows a systematic $\lambda(E)$ trend that explains $\eta^2 = 0.41$–0.43 of the FTLE variance (40–160 bins). The remaining 59% is *within*-shell dispersion — and the horizon sweep shows the total is frozen, so the within-shell component does not self-average either: at fixed energy the phase space is itself mixed (islands, sticky layers, possibly several stochastic components), and the ensemble is a mixture over those invariant structures as well.

Energy bands recapitulate the taxonomy. The per-band FTLE shapes reproduce the entire two-class phenomenology *inside one system*: near the bottom of the release range ($E \approx -18.5$, where chaos is a thin layer) the band is strongly leptokurtic and right-skewed (skew +2.3, excess kurtosis +9.0) — the non-mixed, thin-web signature; mid-range bands ($E \approx -8$ to

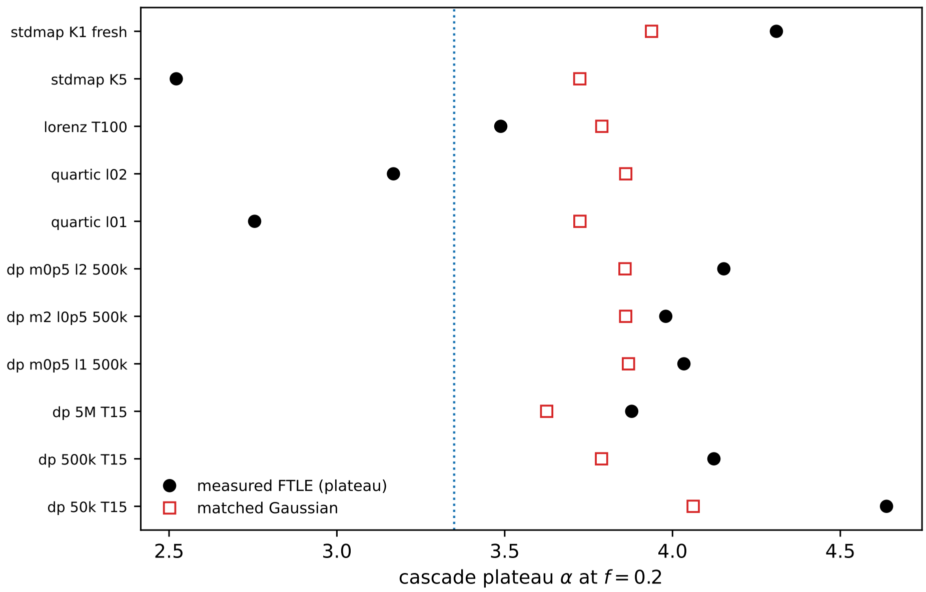

Figure 5: Plateau readings for all eleven ensembles (black) versus their matched-Gaussian nulls (open red). Mixed-phase ensembles sit above their nulls, non-mixed below: 11/11 sign-consistent.

$+2$) are broad and platykurtic (-0.9 to -1.1) — the mixed-phase signature, now manifestly *at fixed energy*, so it is dynamical rather than an artifact of shell mixing; and the highest bands ($E \approx +24$, nearly uniformly chaotic shells) narrow toward Gaussian (cv 0.24, excess kurtosis -0.26) — approaching the self-averaging class. A band-Gaussian mixture reconstruction (λ-band mean plus band-width noise) reproduces the broad platykurtic character only partially (cv 0.61 vs. 0.71), confirming residual within-band structure beyond band-Gaussianity.

The synthesis: *the celebrated double-pendulum constants are frozen statistics of a non-ergodic ensemble* — a mixture over energy shells and, within shells, over coexisting invariant structures. They are reproducible at fixed protocol (hence the hard within-class universality of §5) and drift lawfully with protocol (T, n, f), because they are readings of a mixture's shape through a marginal renormalization flow, not spectral invariants of a single orbit family. The standard-map pair, which has no energy to mix, shows the purely dynamical version of the same dichotomy; the pendulum shows both layers at once.

8 Out-of-sample test: Hénon–Heiles across its transition

The two-class classifier makes a falsifiable prediction: *any* mixed-phase Hamiltonian should present a platykurtic FTLE bulk and read above its matched-Gaussian null, and the Hénon–Heiles system [23] at intermediate energy was named explicitly. We then built the experiment: a Yoshida-4 symplectic, variational-Benettin FTLE engine for $V(x,y) = \frac{1}{2}(x^2+y^2) + x^2y - \frac{1}{3}y^3$ (initial conditions sampled exactly on the energy shell, max $|H-E| = 6\times10^{-17}$; escape-guarded; $n = 2.5 \times 10^5$ per energy, $T = 800$, $\mathrm{d}t = 5 \times 10^{-3}$), sweeping the energy knob that morphs its phase space from integrable-like to almost fully chaotic.

Three regimes emerge (Table 3). *The predicted transition is observed:* at $E = 0.100$ the chaotic component is a thin web (5.7%) and reads *below* its null with a near-Gaussian band shape; at $E = 0.125$, where the classical transition floods half of phase space, the bulk turns platykurtic (excess kurtosis $-0.36 \to -0.81$) and the reading flips *above* the null — the mixed-phase signature switching on exactly where islands and sea begin to coexist at scale. *The*

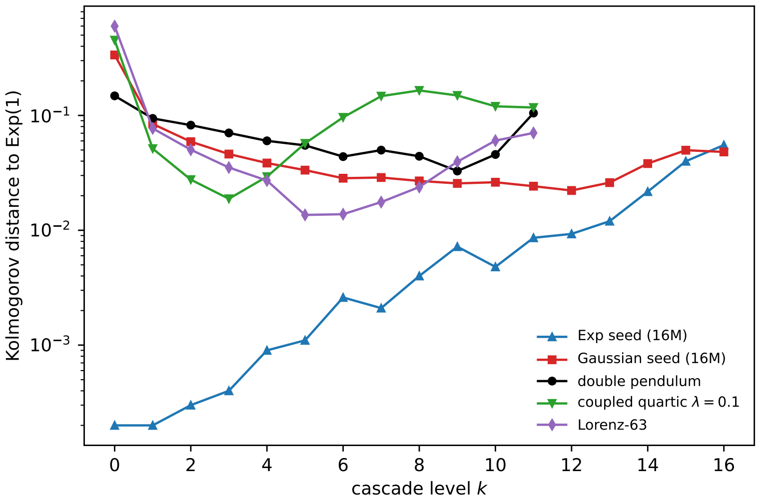

Figure 6: Metastability. Kolmogorov distance to Exp(1) per cascade level for a 1.6×10^7-point exponential seed (blue: sampling noise only), Gaussian seed (red: fast collapse, then a stall at $d \approx 0.02$–0.03 for ≥ 8 levels), and the real systems.

Table 3: Hénon–Heiles energy sweep ($n = 2.5 \times 10^5$ per energy, $T = 800$; chaotic subset at threshold $0.021 \approx 2.5\times$ the regular tangent floor $\ln T/T$). Chaotic fractions track the classical transition. Sign of the matched-null excess in bold.

E	chaotic frac	cv	skew	ex.kurt	α_0	plateau	null	excess
1/24	0.000	—	—	—	—	—	—	—
1/12	0.002	chaotic subset too small for shape statistics						
0.100	0.057	0.235	+0.53	−0.36	9.98	3.752	3.841	**−0.089**
0.125	0.501	0.295	−0.15	−0.81	11.77	4.198	3.924	**+0.274**
0.150	0.816	0.189	−1.39	+2.96	18.65	3.737	3.847	**−0.110**
1/6	0.943	0.198	−1.78	+3.53	23.36	4.201	3.840	**+0.361**

taxonomy's far side also appears: by $E = 0.150$ the sea dominates (82%), the bulk narrows and becomes left-skewed leptokurtic (skew −1.39, excess kurtosis +2.96 — the same signature as the fully chaotic standard map at K=5), and the reading drops back below the null, as self-averaging predicts. *And the sweep exposes a boundary case the two-class taxonomy does not cover:* at the escape energy $E = 1/6$, where the potential's saddles open and trajectories linger near marginally-open channels, the reading flips above the null again despite a leptokurtic bulk — a strong-stickiness regime at the edge of transient chaos [30], flagged here as an open case rather than forced into either class. The level-zero law of Proposition 2 passes its own check en route: at the most Gaussian energy ($E = 0.100$: skew +0.5, kurtosis −0.4) it predicts $\alpha_0 = 10.1$ against the measured 9.98, while the skewed/kurtotic energies deviate exactly where the first-order law should fail. Finally, the sweep retires a documentary loose end of the source program: a claimed "$\alpha_0 = 8.342$ at $E = 0.150$" (already withdrawn in the source's final papers) is not reproducible — we measure 18.65 at this protocol, and §7 shows α_0 is a horizon-indexed bulk-width statistic in any case ($5.6 \to 14.9$ across T for the pendulum), so no protocol-free value of it exists to confirm.

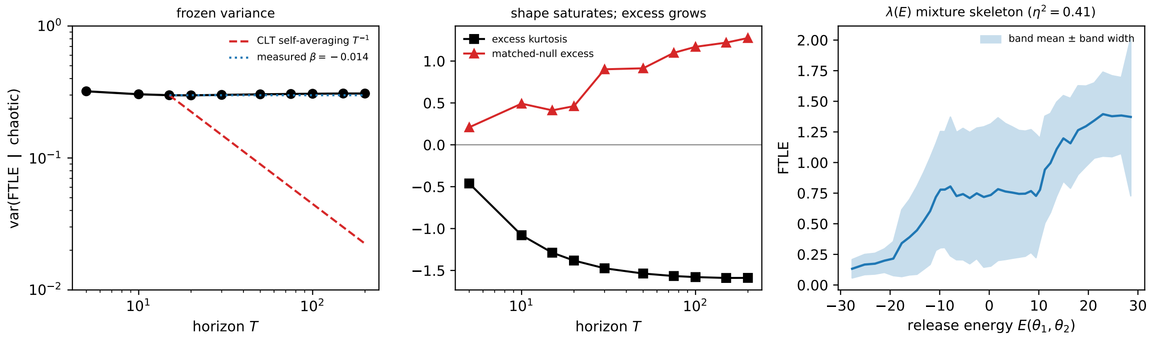

Figure 7: The ensemble structure of the pendulum constants. Left: chaotic-subset FTLE variance across a 13× range of horizons — flat ($\beta = -0.014$), against the CLT's T^{-1}. Middle: the shape saturates (excess kurtosis → −1.59) while the matched-null excess grows. Right: the $\lambda(E)$ mixture skeleton over release energy (band mean ± band width), explaining $\eta^2 = 0.41$ of the variance; the bands themselves span the two-class taxonomy.

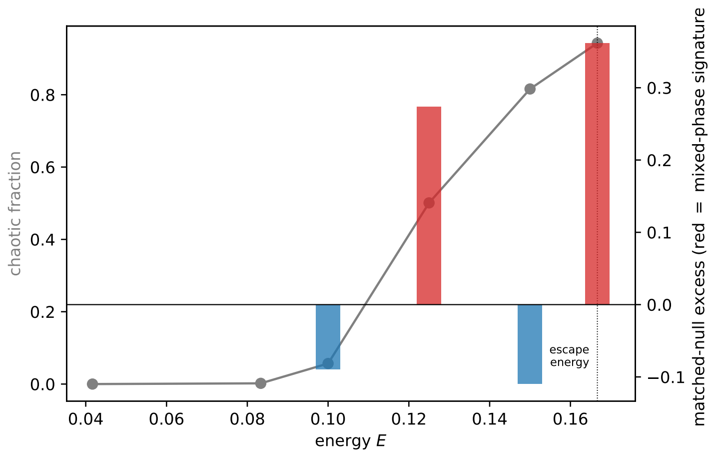

Figure 8: Hénon–Heiles across its transition: chaotic fraction (grey) and matched-null excess (bars; red = positive = mixed-phase signature) versus energy. The excess flips positive exactly where the mixed sea floods phase space ($E = 0.125$), returns negative as chaos dominates ($E = 0.150$), and flips again at the escape energy — the flagged boundary regime.

9 Dynamical mechanism, interpretation, and predictions

Why platykurtic? A mixed phase space at moderate horizon is a *superposition*: the chaotic sea's core produces FTLEs near the ergodic mean, while the hierarchy of sticky boundary layers [7, 8, 10] contributes subpopulations at depressed exponents — trajectories that spent part of the window trapped. Superposing subpopulations with displaced means broadens and *flattens* the mixture (variance mixing drives kurtosis below Gaussian), exactly the cv ≈ 0.6–0.8, excess-kurtosis ≈ −1.2 shape measured here for every mixed-phase ensemble. Absent islands, the mechanism is gone: strongly chaotic or dissipative systems self-average (correlations decay fast; the FTLE is a T-average of a rapidly mixing observable), giving narrow bulks whose residual shape is set by rare fluctuations — hence small cv with skewed, leptokurtic corrections (Lorenz, K=5), while near-regular dynamics piles FTLEs at the regular floor with a thin upper structure (quartic: skew +1.1 to +5.3). Section 7 unifies the picture: in every case the platykurtic broad bulk is the signature of a *non-ergodic ensemble* — a superposition over invariant structures with distinct exponents, whether energy shells (the pendulum's release protocol), islands versus sea (the standard map at $K = 1$), or sticky sublayers within one shell. The FTLE *distribution shape* is thus a direct, cheap probe of the ensemble's ergodic architecture, requiring no phase-space sections (cf. the shape-based predictability program of [28]), and the cascade — with its exactly known exponential reference — is a sensitive one-number readout of that shape. That the *fluctuations* of finite-time exponents, far more than their mean, carry this dynamical information has a clear precedent: Tomsovic and Lakshminarayan showed that the variance and higher cumulants of the finite-time stability exponent in the standard map exhibit anomalous time-scalings that detect regular islands as small as 0.01% of phase space [29]. Our approach is complementary but distinct in three respects: we act on the *entire distribution shape* through an exactly-solvable renormalization operator rather than on low-order cumulant scalings; we obtain a *discrete two-class split* across five distinct systems, with the exponential distribution as an exact separatrix; and we trace the reported "$\alpha^* \approx 4$" constant to a metastable plateau of that operator.

Relation to extreme-value theory. The dichotomy echoes, but is not reducible to, the Fisher–Tippett–Gnedenko trichotomy [17, 18, 21]: the exponential is simultaneously the Gumbel-domain excess law (Pickands–Balkema–de Haan [19, 20]) and the unique fixed point of pair-differencing (Puri–Rubin [14]). The cascade replaces threshold excess by iterated differencing — a renormalization semigroup in the spirit of [22] whose fixed-point structure appears new; its slow manifolds, not its fixed point, carry the physics observed at finite depth.

Predictions — two now tested. The two-class picture makes five falsifiable predictions; two are tested here. (1) *Horizon dependence* — TESTED (§7): the outcome is stronger than predicted. The pendulum ensemble does not merely resist self-averaging; its FTLE variance is frozen outright ($\beta \approx 0$), because the release ensemble is a mixture over invariant structures. The class signature (above-separatrix reading, positive matched-null excess) holds at all ten horizons. (2) *Window signatures* remain open: non-mixed readings track the exponential f-curve (4) (verified here); mixed-phase readings should cross it only at markedly smaller f, when the sticky subpopulations leave the window. (3) *Out-of-sample systems* — TESTED (§8): Hénon–Heiles flips its matched-null excess positive exactly at the thin-web → mixed-sea transition and re-enters the self-averaging class as the sea dominates; one boundary regime (the escape energy) sits outside the two-class taxonomy and is flagged open. (4) *Deeper cascades* remain open: at $n \gg 10^7$ mixed-phase readings should continue the algebraic drift toward 3.3494 measured in §2.7–6; the class *contrast* at matched depth should persist. (5) A *microcanonical* pendulum ensemble (single energy shell) at an energy where the shell is nearly uniformly chaotic should lose the platykurtic signature and read like the non-mixed class — the class membership of an ensemble is set by its ergodic architecture, not by the system's name.

Open mathematical problems. (i) Spectral analysis of T linearized at Exp: our numerics indicate a nearly-degenerate contraction (d_k shrinking $\sim$ 5%/level along the Gaussian manifold); identify the slow eigenfunction and its eigenvalue. (ii) Classification of fixed shapes of $|A - B| \overset{d}{=} cX$ for $c < 1$ (the $c = 1$ case is [14]). (iii) A quantitative map from island-hierarchy exponents (e.g. the universal recurrence decay $z \approx 1.57$ [10]) to the platykurtic FTLE shape and its cascade excess. (iv) The empirical decay constant $r \approx 0.46$–0.51 of the mixed-phase excess: derived, it would make the plateau value itself computable.

10 A number-theoretic audit of the constants

Every constant in this study was swept against a curated library of 67 named mathematical constants (transcendentals and their simple combinations; the Euler–Mascheroni, golden-ratio, Apéry, Catalan, Khinchin and Glaisher–Kinkelin constants; Riemann zeta values; the Feigenbaum and random-matrix-theory families; and KAM/standard-map constants), 159 low-order rationals m/n ($m, n \leq 16$) and the integers up to 1024, together with integer-relation (PSLQ) tests (`mpmath`, maximum coefficient 10^9, 30-digit working precision) on the reported constant triplets, at match tiers $\tau_1/\tau_2/\tau_3 = 0.3\%/0.03\%/0.001\%$; we report the audit so that no numerological reading survives by omission.

- $\eta_{0.2}(\mathrm{Exp}) = 3.3494450\ldots$ has the exact form $1 + [5E_1(\ln 5)]^{-1}$ (eq. 4); the near-miss $10/3$ (0.48%, outside τ_1) is superseded by the closed form. Likewise $\eta_{0.2}(\mathrm{HN}) = 4.4448$ and $\eta_{0.2}(\mathrm{U}) = 9.6417$ are integrals, not numerology.
- The historical conjecture $\alpha^* = 4$ exactly is *excluded*: the flat-plateau parameter scan pins $3.896\,[3.850, 3.942]$ and the 5×10^6 cascade gives $3.710\,[3.675, 3.895]$. The speculative closed form $4 - 1/(4\pi^2) = 3.9747$ (motivated by a legacy pipeline value 3.9762, which it matches to 0.04%) is likewise excluded by both modern intervals; under §6 such n-indexed plateau values are not expected to admit closed forms at all.
- Recurring loose coincidences are logged and rejected: class-mean/4 $= 0.974 \approx K_c = 0.9716$ (0.25%, τ_1; no mechanism, two unrelated scales); noise-floor ratios $0.9241 \approx 1/\zeta(4)$ (0.02%, τ_2; a ratio of two near-equal noise scales); 3/16-like hits on standard errors. PSLQ finds no low-order integer relations among {class means, 4, noise scales} beyond spurious large-coefficient ones.
- The level-zero "constant" $\alpha_0 \approx 8$–8.4 across double-pendulum variants is explained to first order by (5) with the measured platykurtic bulks; its stability across variants reflects the stability of the *shape*, i.e. of the class, not an independent number — and it is horizon-indexed outright ($5.6 \to 14.9$ across $T = 5$–200, §7).
- The source program's decay constant $r \approx 0.46$ "nearly constant across all systems" — and the derived quantity $\rho_2 = \log_2 r \approx -1.12$ on which a claimed mechanistic derivation was built — is the cumulant-halving transient of Proposition 4: $\log_2 r = -1$ exactly at leading order, with computable corrections. No free constant remains.
- A separate near-coincidence from the source program's natural-unit runs — mean chaotic FTLE $0.398821 \approx 1/\sqrt{2\pi} = 0.398942$ (relative 3×10^{-4} at $n = 5 \times 10^6$, and seed-stable to 6×10^{-5} on one integrator arm) — fails constant-hood on the same grounds as everything else in this audit: a second integrator arm shifts it by 0.6%, i.e. it is protocol-indexed. Under §7 such ensemble means are mixture statistics; agreement with a named constant at one protocol point is numerology unless it survives protocol variation, which this does not.
- The relaxation exponent $\theta \approx 0.5$–0.6 of §2.7 is a measured quantity with a conjectured origin (marginal polynomial tower); deriving it exactly is an open problem, and we explicitly do *not* assign it a closed form.

11 Conclusions

A renormalization cascade that arose as an exploratory data-analysis device turns out to possess exact structure: a unique exponential fixed point with a derivable, window-indexed reading (3.3494 at $f = 0.2$); a one-step universal collapse of all Gaussian bulks; an exact cumulant flow under symmetrization (odd cumulants annihilated, standardized even cumulants scaled by 2^{1-m}) that derives the source program's "universal" decay constant $r \approx 1/2$; a closed-form linearized eigenfamily $\mu(s) = 2/(2 - s)$ sitting atop a unit-diagonal polynomial tower — a gapless, non-normal structure whose measured consequence is algebraic relaxation and metastable plateaus; and provable asymptotic tail conservation that finite windows never see. Recast on this foundation, a large-scale computational claim of a new universal constant "$\alpha^* \approx 4$" of double-pendulum chaos resolves into something both humbler and more interesting. The constant is a plateau reading of a marginal flow, taken on an ensemble whose dispersion never self-averages ($\beta \approx 0$ across a $13\times$ range of horizons) because the canonical protocol builds a mixture — over energy shells, and within shells over coexisting invariant structures — whose bands internally recapitulate the very taxonomy found across systems. The universality that survives lives one level up: mixed-phase ensembles — identifiable by broad, flat-topped, platykurtic FTLE bulks — read strictly above the exponential separatrix; fully chaotic, dissipative and near-regular ensembles — narrow or skewed, leptokurtic — read below it, in eleven of eleven archetypal ensembles across five systems, surviving a same-map/same-protocol control, a zero-within-class-heterogeneity meta-analysis, and an out-of-sample Hénon–Heiles sweep that flips sign on cue at the thin-web → mixed-sea transition (and honestly flags one boundary regime, the escape energy, as beyond the taxonomy). The finite-time Lyapunov distribution's shape, read through an operator with an exactly known reference point, is a practical probe of the ergodic architecture of an ensemble — and the exponential distribution, once again, sits at the center of the story.

Acknowledgments. The double-pendulum investigation that grew into this study was sparked by *Drew's Campfire*, and specifically by the video at `https://www.youtube.com/watch?v=8jVogdTJESw&t=64s`, which inspired the entire campaign. I am profoundly grateful to Mr. Varnajoshyula Nagarjuna, my Mathematics: Applications and Interpretations (Higher Level) teacher in the International Baccalaureate — the finest teacher I have had, who first kindled my love of statistical mathematics.

Data and code availability. The cascade operator, the calibration, linearization, ensemble and Hénon–Heiles analysis scripts, the constant-audit library, the raw FTLE ensembles (each with an SHA-256 manifest), and the result files that together reproduce every number and figure in this paper are archived at `https://github.com/RajveerKapoor/lyapunov-cascade-p1/tree/v1.0` (release `v1.0`); a file inventory is given in Appendix B.

A Derivations of the closed-form readings

Exponential. With $u = \ln(1/f)$ and $E \sim \mathrm{Exp}(1)$ (memorylessness),

$$\mathbb{E}\left[\ln\frac{X}{u}\,\middle|\,X > u\right] = \int_0^\infty e^{-t}\ln\!\Big(1+\frac{t}{u}\Big)\,\mathrm{d}t = \int_0^\infty \frac{e^{-t}}{u+t}\,\mathrm{d}t\cdot\Big[\text{by parts}\Big] = e^u E_1(u),$$

giving (4). Values: $E_1(\ln 5) = 0.0851265$, so $\eta_{0.2} = 1+(5\times 0.0851265)^{-1} = 3.3494450$; similarly $\eta_{0.1} = 4.0873926$, $\eta_{0.3} = 2.9057918$, $\eta_{0.05} = 4.8111467$.

Half-normal (the universal Gaussian level-1). $u = \Phi^{-1}(0.9) = 1.281552$ (the 80th percentile of $|N(0,1)|$), and $\eta_{0.2} = 1 + \big[\frac{1}{0.2}\int_u^\infty \ln(x/u)\,2\varphi(x)\,\mathrm{d}x\big]^{-1} = 4.444794$ by numerical quadrature.

Uniform. $u = 0.8$: $\mathbb{E}[\ln(X/0.8) \mid X > 0.8] = 5\int_{0.8}^1 \ln(x/0.8)\,\mathrm{d}x = 0.115780$, so $\eta_{0.2} = 9.641716$. (Bounded supports read high: the log-excess is compressed.)

Gaussian level-zero law. For $X = |N(\mu,\sigma^2)|$, $\mathrm{cv} = \sigma/\mu$ small: $u = \mu + \sigma z_f$ and $\mathbb{E}[X - u \mid X > u] = \sigma m_f$ with $m_f = \varphi(z_f)/f - z_f$; expanding $\ln(x/u) \approx (x-u)/u$ gives (5). Second-order corrections are $O(\mathrm{cv})$ relative and account for the observed 3–5% underprediction.

B Data provenance and reproducibility

- Double pendulum: `lyapunov_all_50k.npy` ($n = 5 \times 10^4$), `ftle_n500k_lagrangian_T15_wave0c.npy`, `ftle_n5M_lagrangian_T15.npy`; geometric variants `disc067_ftle_dp_{m0p5_l1, m2_l0p5, m0p5_l2}_n500k_T15.npy`.
- Standard map: `standard_map_K5_ftle_n500k.npy`; the $K = 1$ ensemble regenerated for this work ($n = 5 \times 10^5$, 10^3 iterations, seed `0xADD`; its cascade fit 4.11/plateau 4.31 agrees with the program's banked 4.382 $[3.646, 4.934]$).
- Lorenz-63: `lorenz_ftle_n500k_T100.npy`. Coupled quartic: `coupled_quartic_ftle_n500k.npy` ($\lambda = 0.1$), `coupled_quartic_lam0p2_ftle_n500k.npy`, with manifests recording integrator, seeds, moments, SHA-256.
- Meta-analytic quantities quoted in §5 (random-effects τ^2, Cochran Q, cross-n correlation, ridge noise-null) were computed from the same production arrays; the corresponding analysis scripts accompany this manuscript as supplementary material.
- Synthetic calibration: 17 families, $n = 5 \times 10^5$ (deep runs 1.6×10^7), seeds `0xF1BE/0xADD/7`; bootstrap-through-the-tree $B = 250$; full results in `synthetic_battery_results.json` and `addendum_results.json`.
- Horizon sweep (§7): `disc024_ftle_T{5..200}_N50000.npy` (canonical protocol re-runs at ten horizons, seed 42); release-angle grid `lyapunov_grid_100x100.npy` (legacy lineage; internal contrasts only). Analyses: `tsweep_analysis.py`, `energy_mixture.py` → `tsweep_results.json`, `energy_mixture_results.json`.
- Linearization (§2.7): `linearization.py` → `linearization_results.json` (cumulant identities on data, eigenfamily and polynomial-tower checks at 900 grid points, uniform-seed 8×10^6 relaxation).
- Hénon–Heiles (§8): `hh_sweep.py` → `hh_sweep_results.json` and `hh_ftle_E*.npy` ($n = 2.5 \times 10^5$ per energy, $T = 800$, Yoshida-4 variational Benettin, on-shell ICs to 6×10^{-17}, escape-guarded; seed `0x11E5`).

C Full synthetic calibration table

D Estimator and extrapolation pathologies

Two failure modes of the published pipeline are documented so that readers do not mistake them for physics. (i) *Flat-sequence degeneracy:* the three-parameter geometric fit $\alpha_k = \alpha^* + Cr^k$ is unidentifiable when the sequence is flat (the exponential seed — its own fixed point): the optimizer runs to its bounds ($\alpha^* \to 12$, $r \to 0.99$) unless enough levels constrain it (≥ 14 levels at $n = 5\times10^6$ recover 3.3507). (ii) *Huge level-zero values:* when α_0 exceeds the fit's initialization envelope (narrow bulks: Lorenz $\alpha_0 = 80$, K=5 $\alpha_0 = 53$, $|N(1, 0.1)|$ $\alpha_0 = 21$), the default fit fails or runs away; dropping level 0 restores a well-posed fit. The fit-free plateau statistic (median of late well-sampled levels) is immune to both and is reported throughout alongside fit values. Deep-level readings carry a small negative finite-n bias (even the exponential's plateau reads 3.30–3.33 at levels with $n_k \lesssim 10^3$), common to all ensembles compared at matched depth.

Conflicts of Interest

The author declares no conflicts of interest.

Table 4: Synthetic battery at $n = 5 \times 10^5$, $f = 0.2$. Plateau = median of levels 2–6. Fit values marked ‡ ran to a parameter bound (Appendix D).

family	EVT domain	α_0	plateau	fit
Exp(1)	Gumbel	3.36	3.34	12.0‡
Exp(1) (5M)	Gumbel	3.35	3.35	3.351
half-normal	Gumbel	4.45	3.79	3.77
$\|N(3,1)\|$	Gumbel	8.64	3.89	3.87
$\|N(1,0.1)\|$	Gumbel	21.3	3.86	3.97*
lognormal σ=0.5	Gumbel	4.59	3.26	3.11
lognormal σ=1	Gumbel	2.79	2.70	7.9‡
Weibull k=1.5	Gumbel	4.52	3.68	1.6‡
Weibull k=0.7	Gumbel	2.64	3.01	3.33
Gamma(3)	Gumbel	4.77	3.58	3.58
uniform$(0,1)$	Weibull	9.61	4.31	4.16
Pareto a=2.5	Fréchet	3.50	2.42	2.34
Pareto a=4	Fréchet	5.01	2.72	2.59
Pareto a=6	Fréchet	6.99	2.91	2.91
bulk + Pareto a=3 (12%)	Fréchet	3.10	2.66	2.27
bulk + Pareto a=4 (12%)	Fréchet	3.35	3.03	2.49
bulk + Pareto a=5 (12%)	Fréchet	3.53	3.28	2.27

*level-0 dropped before fitting.

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